Gas Laws

61. (d) $\frac{P_1V_1}{T_1}=\frac{P_2V_2}{T_2}\Rightarrow T_2=\frac{P_2V_2}{P_1V_1}\times T_1$

$\Rightarrow T_2=\frac{1}{30}\times\frac{10}{1}\times 300=100K=-173°C$

62. (c) As temperature decreases to half and volume made twice, hence pressure becomes $\frac{1}{4}$ times.

63. (d) $\frac{P_1V_1}{T_1}=\frac{P_2V_2}{T_2}\Rightarrow V_2=\frac{P_1V_1}{T_1}\times\frac{T_2}{P_2}$

$\Rightarrow V_2=\frac{1500\times 4\times 270}{300\times 2}=2700m^3$

64. (d)

65. (c) $\frac{P_1V_1}{T_1}=\frac{P_2V_2}{T_2}$

$T_2=\frac{P_2V_2}{P_1V_1}T_1=\frac{2}{1}\times\frac{3}{1}\times 300=1800K=1527°C$

66. (b) $PV=mrT=m\left(\frac{R}{M}\right)T=\frac{8}{32}RT$ [$\because M_{O_2}=32$]

$\Rightarrow PV=\frac{RT}{4}$.

67. (a) $PV=\mu RT=\frac{m}{M}RT=\frac{5}{32}RT$.

68. (d) $PV=\mu RT\Rightarrow P\left(\frac{m}{\rho}\right)=\mu RT\Rightarrow \rho\propto\frac{T}{P}$

Since T becomes twice and P becomes four times so ρ becomes $\frac{1}{2}$ times.

69. (d) $V\propto T\Rightarrow\frac{V_1}{V_2}=\frac{T_1}{T_2}\Rightarrow\frac{V}{2V}=\frac{(273+0)}{T_2}$

$\Rightarrow T_2=546K=273°C$

70. (c) $\mu_1=\frac{PV}{RT},\ \mu_2=\frac{PV}{RT}$

$$P' = \frac{(\mu_1 + \mu_2)RT}{V} = \frac{2PV}{RT} \times \frac{RT}{V} = 2P$$

71. (c) According to the Dalton's law of partial pressures, the total pressure will be $P_1 + P_2 + P_3$.

72. (c) $PV = mrT \Rightarrow P \propto m$, ($V$, $T \rightarrow$ constant)

73. (d) It is because of their low densities.

74. (b) $PV = mrT = P \propto mT \Rightarrow \frac{P_1}{P_2} = \frac{m_1}{m_2} \times \frac{T_1}{T_2}$

$$\Rightarrow \frac{10}{P_2} = \frac{m}{m/2} \times \frac{(273+27)}{(273+87)} \Rightarrow P_2 = 6\,atm$$

75. (c) $P \propto T \Rightarrow \frac{P_1}{P_2} = \frac{T_1}{T_2} \Rightarrow \frac{P_1}{(1.01)P_1} = \frac{T_1}{T_1 + 5}$

$\Rightarrow T_1 = 500\,K = 227°C$.

76. (c) The number of moles of the system remains same,

$$\frac{P_1V_1}{RT_1} + \frac{P_2V_2}{RT_2} = \frac{P(V_1+V_2)}{RT} \Rightarrow T = \frac{P(V_1+V_2)T_1T_2}{(P_1V_1T_2 + P_2V_2T_1)}$$

According to Boyle's law,

$$P_1V_1 + P_2V_2 = P(V_1+V_2) \quad \therefore T = \frac{(P_1V_1 + P_2V_2)T_1T_2}{(P_1V_1T_2 + P_2V_2T_1)}$$

77. (c)

78. (a)

79. (c) In Vander Wall's equation $\left(P + \frac{a}{V^2}\right)(V-b) = RT$

a represents intermolecular attractive force and *b* represents volume correction

80. (c) $PV = \mu RT = \frac{m}{M}RT \Rightarrow P \propto mT$

$$\Rightarrow \frac{P_2}{P_1} = \frac{m_2}{m_1}\frac{T_2}{T_1} = \frac{1}{2} \times \frac{(273+27+50)}{(273+27)} = \frac{7}{12}$$

$$\Rightarrow P_2 = \frac{7}{12}P_1 = \frac{7}{12} \times 20 = 11.67\,atm \approx 11.7\,atm$$